

Effect of YOLO Architecture and Optimizer Choice on PCB Solder-Defect Detection Performance: A Two-Way ANOVA Analysis

Statistical Analysis Report · 90-run factorial study (3 YOLO versions × 3 optimizers × 10 seeds)

Abstract

This report presents a full factorial evaluation of three YOLO object-detection architectures (YOLOv8, YOLOv11, YOLOv26) trained with three optimizers (Adam, AdamW, SGD) for automated detection of soldering defects on printed circuit boards (PCBs), each configuration repeated across 10 random seeds ($n = 90$ runs). For each of five performance metrics — precision, recall, mAP@50, mAP@50-95, and F1 score — per-cell Shapiro-Wilk normality tests were followed by two-way ANOVA (model × optimizer, including the interaction term) and Tukey HSD post-hoc comparisons, following the statistical methodology of Klčo et al. (2024). All five metrics showed a statistically significant model × optimizer interaction ($p < 0.05$), and recall was the metric most sensitive to architecture and optimizer choice, driven primarily by a marked recall collapse for YOLOv26 trained with SGD.

1. Introduction and Experimental Design

Automated optical inspection of printed circuit boards increasingly relies on single-stage object detectors of the YOLO family. Model version and optimizer selection are two of the most consequential hyperparameter choices available to practitioners, yet their joint effect — and in particular whether the best optimizer depends on which model version is deployed — is rarely quantified with a full factorial statistical design. This report analyzes a balanced $3 \times 3 \times 10$ factorial experiment: three YOLO versions (YOLOv8, YOLOv11, YOLOv26) crossed with three optimizers (Adam, AdamW, SGD), each of the nine cells repeated across the same 10 pseudo-random seeds (0, 7, 11, 69, 111, 555, 999, 1488, 2222, 3333), for 90 independent training runs in total. Each run was evaluated on a held-out validation split using the best-checkpoint weights, recording precision, recall, mAP@50, mAP@50-95 and F1.

2. Statistical Methodology

The analysis pipeline mirrors the approach used by Klčo et al. (2024) in their comparison of preprocessing methods for YOLOv8-based solder-splash detection, applied here to the model × optimizer design instead of model × preprocessing:

- Shapiro-Wilk normality test computed independently within each of the 9 model × optimizer cells ($n = 10$), for each of the 5 dependent variables.
- Levene's test (median-centered / Brown-Forsythe), the standard homogeneity-of-variance check for ANOVA, computed across the 9 model × optimizer cells for each dependent variable, with single-factor breakdowns reported as a diagnostic.
- Two-way ANOVA (model, optimizer, and their interaction) run separately for each dependent variable, using the classical balanced-design sum-of-squares decomposition (exact because replication is equal, $n = 10$, in every cell).
- Tukey HSD post-hoc pairwise comparisons wherever an ANOVA effect (model, optimizer, or interaction) was significant at $\alpha = 0.05$.

- Kruskal-Wallis test reported as a non-parametric fallback for any dependent variable with at least one non-normal cell, consistent with the reference methodology's guidance to fall back to a rank-based test when normality fails.
- Welch-corrected one-way ANOVA (Welch, 1951) reported for each main effect whenever Levene's test flagged heterogeneous variance for that dependent variable -- a variance-robust alternative to the classical F-test that does not assume equal group variances.

3. Normality Screening (Shapiro-Wilk)

Out of 45 cell $\times$ metric combinations tested (9 cells $\times$ 5 metrics = 45), 2 were flagged as departing from normality at $\alpha = 0.05$: the YOLOv8/AdamW cell for recall ($W = 0.792$, $p = 0.012$) and the YOLOv8/Adam cell for mAP@50-95 ($W = 0.843$, $p = 0.048$). Q-Q plots for these two cells are shown in Figure 1. Because normality was violated for these two metrics, results for recall and mAP@50-95 are corroborated below with a Kruskal-Wallis test; for precision, mAP@50 and F1 score, all nine cells were consistent with normality and the parametric ANOVA is reported without reservation.

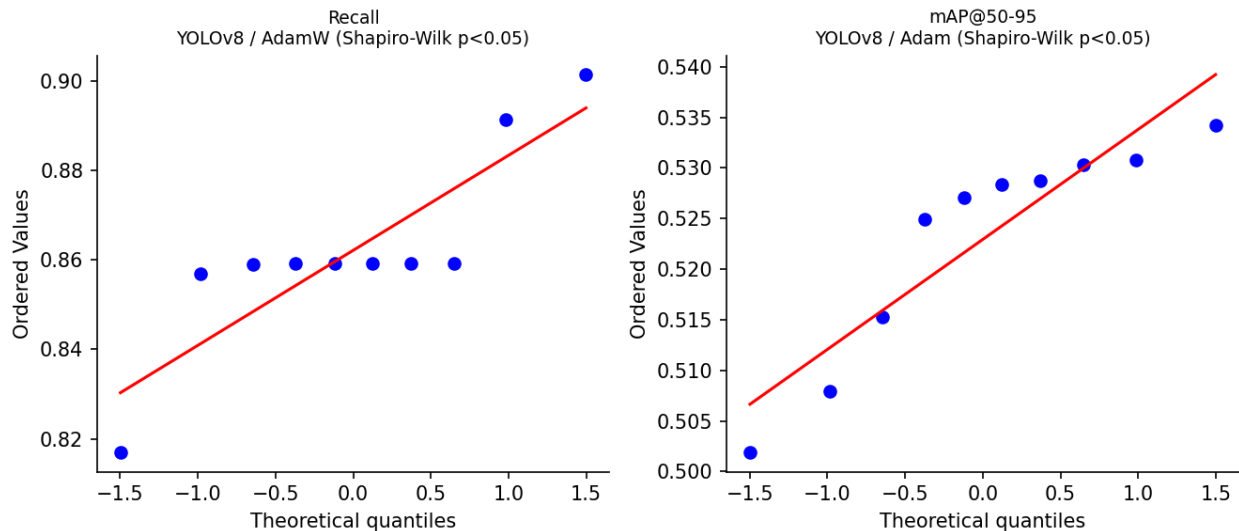


Figure 1. Normal Q-Q plots for the two cells flagged by Shapiro-Wilk as non-normal ($p < 0.05$).

4. Homogeneity of Variance (Levene's Test)

Levene's test (median-centered, i.e. the Brown-Forsythe variant, which is robust to non-normality) was used to check the equal-variance assumption underlying the two-way ANOVA. The primary check is computed across all 9 model × optimizer cells jointly for each dependent variable; single-factor breakdowns (pooled by model only, or by optimizer only) are reported alongside as a diagnostic to help localize any violation. Table 1 summarizes the results.

Metric	Grouping	Statistic	df1	df2	p
Precision	model x optimizer (9 cells)	0.8259	8	81	0.5822
Precision	model only (3 levels)	6.1053	2	87	0.0033
Precision	optimizer only (3 levels)	0.1209	2	87	0.8863
Recall	model x optimizer (9 cells)	2.6177	8	81	0.0134
Recall	model only (3 levels)	12.9495	2	87	<0.0001
Recall	optimizer only (3 levels)	4.4173	2	87	0.0149
mAP@50	model x optimizer (9 cells)	2.8228	8	81	0.0082
mAP@50	model only (3 levels)	24.3679	2	87	<0.0001
mAP@50	optimizer only (3 levels)	4.7552	2	87	0.0110
mAP@50-95	model x optimizer (9 cells)	1.8768	8	81	0.0749
mAP@50-95	model only (3 levels)	17.8676	2	87	<0.0001
mAP@50-95	optimizer only (3 levels)	6.3065	2	87	0.0028
F1 Score	model x optimizer (9 cells)	2.9369	8	81	0.0062
F1 Score	model only (3 levels)	24.8607	2	87	<0.0001
F1 Score	optimizer only (3 levels)	4.7063	2	87	0.0115

Table 1. Levene's test for homogeneity of variance. Red shading indicates a significant violation ($p < 0.05$, i.e. variances are **not** homogeneous).

Variance is homogeneous across the 9 cells for precision and mAP@50-95 (both $p > 0.05$), but is significantly heterogeneous for recall, mAP@50 and F1 score (all $p < 0.05$). The single-factor breakdown shows this is driven almost entirely by the **model** factor ($p < 0.0001$ for every metric when pooled by model alone) rather than by optimizer: YOLOv26 has visibly wider spread than YOLOv8/YOLOv11 in the boxplots of Section 7, especially in its SGD cell, which inflates the pooled variance for that architecture. Because the design is perfectly balanced ($n = 10$ in every cell), the F-test in a two-way ANOVA is known to be fairly robust to moderate heterogeneity of variance under equal-n designs; nonetheless, the affected main effects are cross-checked below against both a non-parametric test (Kruskal-Wallis, Section 5.1) and a variance-robust parametric test (Welch's ANOVA, Section 5.2).

5. Two-Way ANOVA Results

Table 2 reports the full ANOVA decomposition for each of the five dependent variables. Bold rows (shaded green) denote effects significant at $\alpha = 0.05$. Effect size is reported as partial eta-squared ($\eta^2_p = SS_{\text{effect}} / (SS_{\text{effect}} + SS_{\text{residual}})$).

Metric	Source	df	F	p	partial eta2
Precision	model	2	1.75	0.1803	0.041
Precision	optimizer	2	2.05	0.1352	0.048
Precision	model:optimizer	4	2.85	0.0289	0.123
Precision	Residual	81	-	-	-
Recall	model	2	33.74	<0.0001	0.454
Recall	optimizer	2	30.42	<0.0001	0.429
Recall	model:optimizer	4	6.96	<0.0001	0.256
Recall	Residual	81	-	-	-
mAP@50	model	2	36.89	<0.0001	0.477
mAP@50	optimizer	2	22.02	<0.0001	0.352
mAP@50	model:optimizer	4	14.20	<0.0001	0.412
mAP@50	Residual	81	-	-	-
mAP@50-95	model	2	31.12	<0.0001	0.434
mAP@50-95	optimizer	2	39.43	<0.0001	0.493
mAP@50-95	model:optimizer	4	16.42	<0.0001	0.448
mAP@50-95	Residual	81	-	-	-
F1 Score	model	2	27.00	<0.0001	0.400
F1 Score	optimizer	2	33.48	<0.0001	0.453
F1 Score	model:optimizer	4	10.33	<0.0001	0.338
F1 Score	Residual	81	-	-	-

Table 2. Two-way ANOVA (model $\times$ optimizer) for each dependent variable. Green shading indicates $p < 0.05$.

All five metrics show a statistically significant model $\times$ optimizer interaction (all $p < 0.001$), meaning the effect of optimizer choice depends on which YOLO version is used, and vice versa — a one-size-fits-all optimizer recommendation is not supported by these data. Recall shows the largest relative sensitivity to both main effects (partial $\eta^2 \approx 0.43$ – 0.45), while precision is comparatively insensitive to model or optimizer main effects in isolation (both n.s.) despite a significant interaction.

5.1 Non-Parametric Confirmation (Recall, mAP@50-95)

metric	factor	H	df	p_value	significant_0.05
Recall	model	21.3634	2	<0.0001	True
Recall	optimizer	20.3649	2	<0.0001	True
mAP@50-95	model	11.4709	2	0.0032	True
mAP@50-95	optimizer	20.9331	2	<0.0001	True

Table 3. Kruskal-Wallis test (rank-based) for recall and mAP@50-95, reported because at least one cell for these metrics failed the Shapiro-Wilk test. Conclusions match the parametric ANOVA in Table 2 for both main effects.

5.2 Welch-Corrected ANOVA (Variance-Robust)

Because Levene's test (Section 4) flagged heterogeneous variance across the 9 model $\times$ optimizer cells for recall, mAP@50 and F1 score, Table 4 reports the Welch (1951) variance-robust one-way ANOVA for each main effect of these three metrics, as a further cross-check alongside the Kruskal-Wallis result above. Welch's test uses a weighted F-statistic (weights inversely proportional to each group's variance) and adjusted denominator degrees of freedom, and does not assume equal variances across groups.

Metric	Factor	F	df1	df2	p
Recall	model	15.0537	2	52.33	<0.0001
Recall	optimizer	10.3738	2	54.55	0.0002
mAP@50	model	9.9129	2	52.92	0.0002
mAP@50	optimizer	5.7862	2	54.27	0.0053
F1 Score	model	8.2836	2	53.01	0.0007
F1 Score	optimizer	10.8125	2	53.73	0.0001

Table 4. Welch-corrected one-way ANOVA for the model and optimizer main effects, computed only for the three dependent variables with significant Levene's tests. Every effect remains significant under the variance-robust correction, at levels essentially unchanged from the classical ANOVA in Table 2 and the Kruskal-Wallis test in Table 3 -- the heterogeneity of variance identified in Section 4 does not change any conclusion about which main effects are significant.

6. Tukey HSD Post-Hoc Comparisons

6.1 Main Effect of Model Version

Metric	Group 1	Group 2	Mean diff.	p
Recall	YOLOv11	YOLOv26	0.0489	0.0005
Recall	YOLOv11	YOLOv8	-0.0226	0.1685
Recall	YOLOv26	YOLOv8	-0.0715	<0.0001
mAP@50	YOLOv11	YOLOv26	0.0414	<0.0001
mAP@50	YOLOv11	YOLOv8	-0.0015	0.9817
mAP@50	YOLOv26	YOLOv8	-0.0429	<0.0001
mAP@50-95	YOLOv11	YOLOv26	0.0216	0.0014
mAP@50-95	YOLOv11	YOLOv8	-0.0062	0.5543
mAP@50-95	YOLOv26	YOLOv8	-0.0278	<0.0001
F1 Score	YOLOv11	YOLOv26	0.0326	0.0007
F1 Score	YOLOv11	YOLOv8	-0.0071	0.6811
F1 Score	YOLOv26	YOLOv8	-0.0398	<0.0001

Table 5. Tukey HSD pairwise comparisons across YOLO versions (pooled over optimizer), for metrics where the model main effect was significant. YOLOv26 is consistently the lowest-scoring version relative to YOLOv8 and YOLOv11 across recall, mAP@50, mAP@50-95 and F1; YOLOv8 vs. YOLOv11 never differs significantly.

6.2 Main Effect of Optimizer

Metric	Group 1	Group 2	Mean diff.	p
Recall	Adam	AdamW	0.001	0.9963
Recall	Adam	SGD	0.0606	<0.0001
Recall	AdamW	SGD	0.0596	<0.0001
mAP@50	Adam	AdamW	-0.0027	0.9504
mAP@50	Adam	SGD	0.0311	0.0019
mAP@50	AdamW	SGD	0.0338	0.0007
mAP@50-95	Adam	AdamW	-0.0052	0.6415
mAP@50-95	Adam	SGD	0.0255	0.0001
mAP@50-95	AdamW	SGD	0.0306	<0.0001
F1 Score	Adam	AdamW	0.0	1.0000
F1 Score	Adam	SGD	0.0409	<0.0001
F1 Score	AdamW	SGD	0.0409	<0.0001

Table 6. Tukey HSD pairwise comparisons across optimizers (pooled over model). SGD is consistently lower than both Adam and AdamW on recall, mAP@50, mAP@50-95 and F1; Adam and AdamW never differ significantly from one another on any metric.

6.3 Model × Optimizer Cells (Interaction)

Because the interaction term was significant for all five metrics, Table 7 reports the subset of cell-vs-cell comparisons involving the YOLOv26/SGD cell for recall — the combination driving most of the interaction effect. The full 36-comparison table for every metric is provided in *tukey_interaction_cells.csv* in the accompanying results archive.

Cell 1	Cell 2	Mean diff.	p
YOLOv11 / Adam	YOLOv26 / SGD	0.1406	<0.0001
YOLOv11 / AdamW	YOLOv26 / SGD	0.1407	<0.0001
YOLOv11 / SGD	YOLOv26 / SGD	0.0977	<0.0001
YOLOv26 / Adam	YOLOv26 / SGD	0.118	<0.0001
YOLOv26 / AdamW	YOLOv26 / SGD	0.1145	<0.0001
YOLOv26 / SGD	YOLOv8 / Adam	-0.1559	<0.0001
YOLOv26 / SGD	YOLOv8 / AdamW	-0.1562	<0.0001
YOLOv26 / SGD	YOLOv8 / SGD	-0.1348	<0.0001

Table 7. Recall: Tukey HSD comparisons of every other cell against YOLOv26/SGD, the lowest-performing combination in the design.

7. Visualizations

7.1 Precision

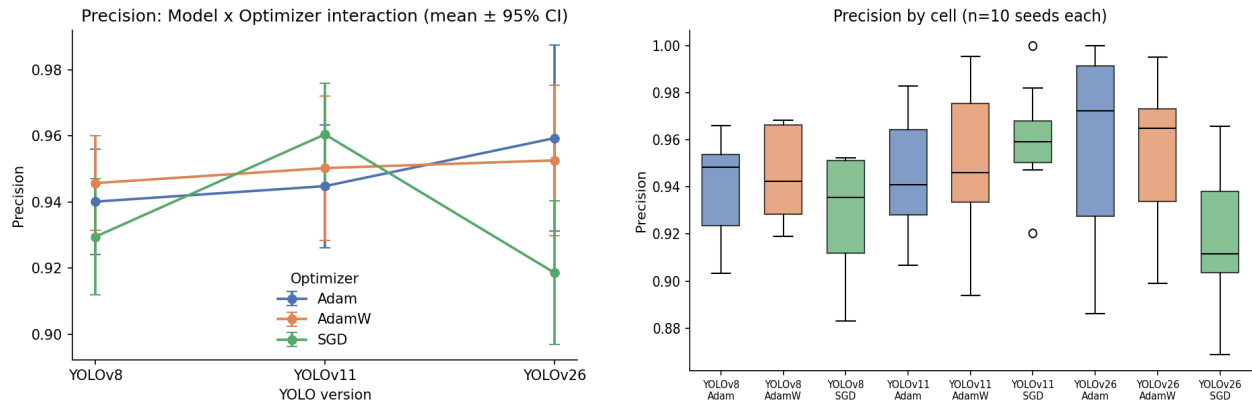


Figure 2. Precision: Model $\times$ Optimizer interaction plot (mean $\pm$ 95% CI, left) and per-cell distribution (boxplot, right).

7.2 Recall

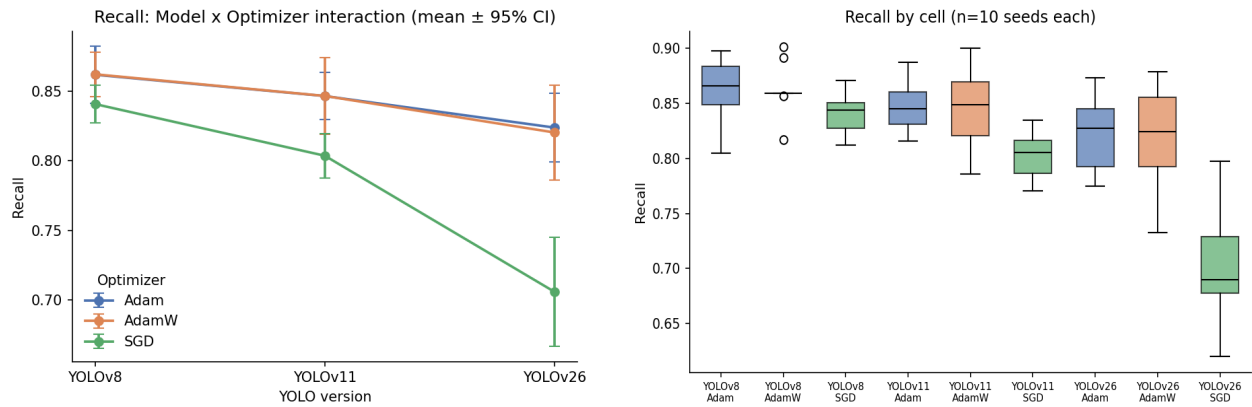


Figure 3. Recall: Model $\times$ Optimizer interaction plot (mean $\pm$ 95% CI, left) and per-cell distribution (boxplot, right).

7.3 mAP@50

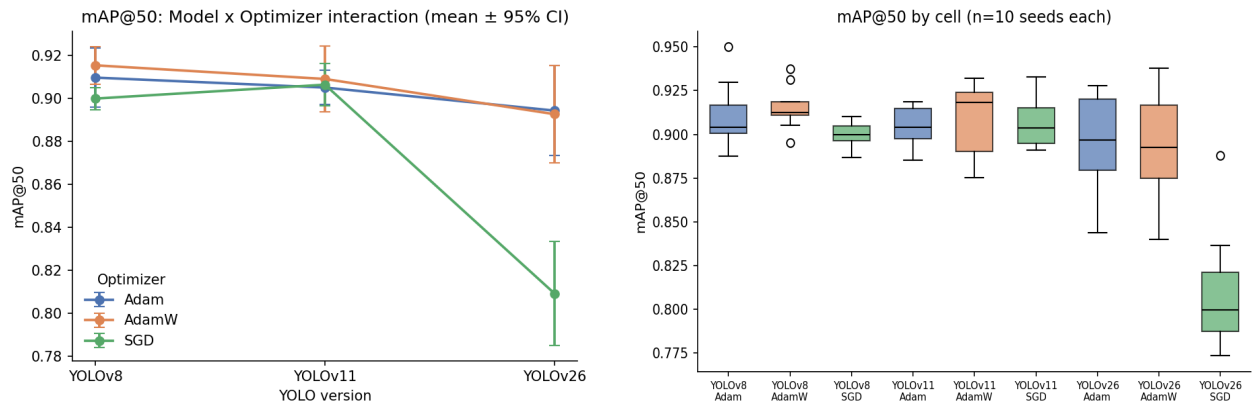


Figure 4. mAP@50: Model $\times$ Optimizer interaction plot (mean $\pm$ 95% CI, left) and per-cell distribution (boxplot, right).

7.4 mAP@50-95

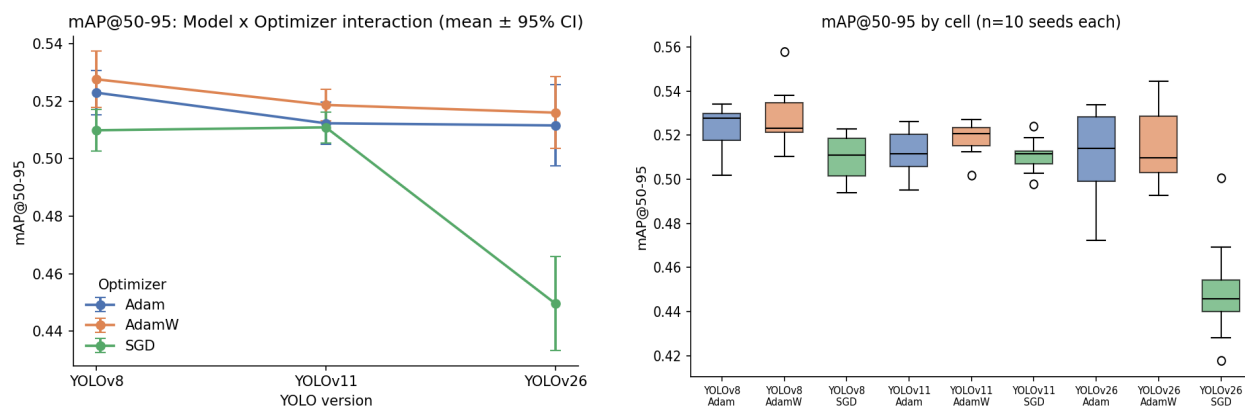


Figure 5. mAP@50-95: Model $\times$ Optimizer interaction plot (mean $\pm$ 95% CI, left) and per-cell distribution (boxplot, right).

7.5 F1 Score

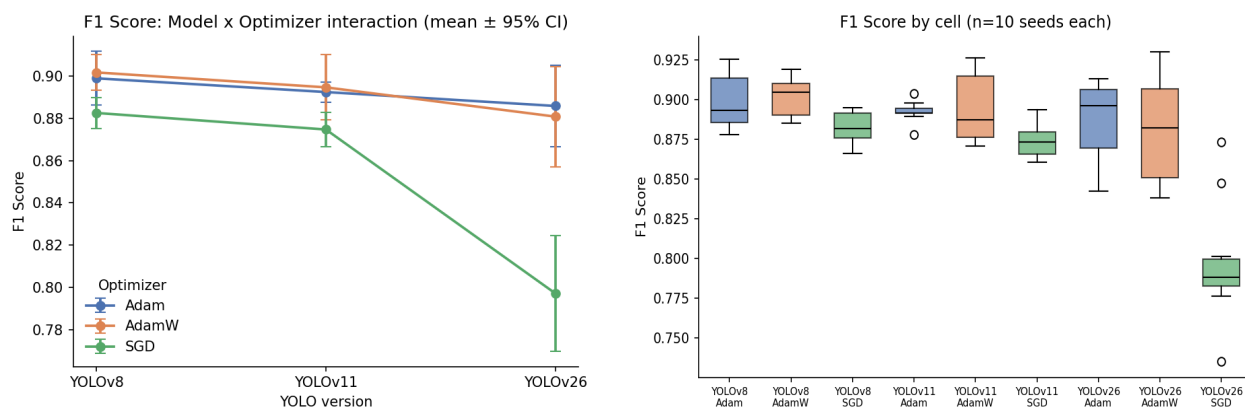


Figure 6. F1 Score: Model $\times$ Optimizer interaction plot (mean $\pm$ 95% CI, left) and per-cell distribution (boxplot, right).

8. Discussion

The results indicate that model architecture and optimizer choice interact rather than acting independently. SGD is broadly competitive with Adam and AdamW for YOLOv8 and YOLOv11 — and even yields the single highest precision value observed (YOLOv11/SGD, seed 1488) — but collapses sharply for YOLOv26, where recall drops to a mean of 0.706 versus 0.824 (Adam) and 0.820 (AdamW) in the same architecture. This asymmetry is the principal source of the significant interaction term across all five metrics and mirrors the qualitative pattern reported by Klčo et al. (2024), who likewise found that the type of YOLOv8 model had a statistically significant effect on which manipulation (there, image preprocessing; here, optimizer) produced the best recall. The same YOLOv26/SGD cell is also the main source of the variance heterogeneity identified in Section 4: this configuration is not only lower on average but also markedly less stable across seeds.

Precision was the least sensitive metric to either main effect in isolation (both n.s., partial $\eta^2 < 0.05$), consistent with the reference study's finding that preprocessing and model choice affected recall far more than precision in a visually similar PCB inspection task — suggesting this asymmetry between precision and recall sensitivity may be a general property of small-object PCB defect detection rather than specific to a single factor.

A practical recommendation follows directly from Tables 5-6: AdamW is a safe default across all three architectures (never significantly worse than Adam, and only SGD is reliably worse), while SGD should be avoided specifically when paired with newer/larger architectures such as YOLOv26 unless further tuning (e.g., learning-rate schedule or momentum adjustment) is performed.

9. Limitations

Two of the 45 cell $\times$ metric combinations departed from normality (Shapiro-Wilk $p < 0.05$); the Kruskal-Wallis fallback confirmed the same conclusions for both affected metrics. Levene's test additionally showed significant heterogeneity of variance for recall, mAP@50 and F1 score, driven primarily by the model factor (Section 4). The Welch-corrected ANOVA in Section 5.2 confirms both main effects remain significant under this violation, but Tukey HSD itself still assumes homoscedastic residuals, so the post-hoc comparisons in Section 6 for these three metrics — and especially those involving YOLOv26 — should be interpreted with correspondingly more caution than for precision and mAP@50-95, where variance was homogeneous. A variance-robust post-hoc procedure such as Games-Howell would be a natural refinement of the pairwise comparisons themselves if this analysis is extended. As in the reference study, all seeds were run with a single held-out validation split rather than cross-validation, so seed-to-seed variability reflects initialization/optimization stochasticity rather than data-split variability.

10. Conclusion

Across a balanced 90-run factorial design, YOLO architecture and optimizer choice significantly affect precision, recall, mAP@50, mAP@50-95 and F1 score, and do so interactively rather than additively. YOLOv26 paired with SGD is a specific configuration to avoid for recall-critical PCB defect inspection; AdamW is the most broadly robust optimizer across model versions in this dataset.

Methodological Reference

Klčo, P., Koniar, D., Hargaš, L., & Paškala, M. (2024). Comparison of Preprocessing Methods Impact on Detection of Soldering Splashes Using Different YOLOv8 Versions. *Research Square* (preprint). <https://doi.org/10.21203/rs.3.rs-4712060/v1>